THE EQUIVARIANT STOLZ EXACT SEQUENCE DESCENDS TO THE QUOTIENT BY THE FUNDAMENTAL GROUP

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Allowed standard prerequisites (editorial).

- The G-equivariant and G-tilde-equivariant Stolz exact sequences attached to X and its covering X-tilde. Set up in the paragraphs preceding the counted statement, ending at source line 509. Fixes the two objects whose isomorphism is asserted and the notion of equivariant cycle and bordism used in the proof.
- The covering X-tilde of X with group of deck transformations pi_1(X) inside G-tilde. Used inside the counted proof at lines 516-529. Supplies the quotient and pull-back constructions that give the two inverse correspondences.

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Proposition 0.1. *The G-equivariant Stolz exact sequence associated to X is isomorphic to the $\widetilde{G}$-equivariant Stolz exact sequence associated to $\tilde{X}$.*

Proof. The action of $\pi_1(X) \subset \widetilde{G}$ on $\tilde{X}$ is free. Consequently, for every $\widetilde{G}$-map $f\colon \bar{M} \to \tilde{X}$ the action of $\pi_1(X)$ on $\bar{M}$ is free. We can therefore quotient out this action and obtain a cycle $M := \bar{M}/\pi_1(X) \to X$ with residual action of $G := \widetilde{G}/\pi_1(X)$. Vice versa, given a G-map $f\colon M \to X$ we can pull back $\tilde{X} \to X$ along f and obtain a $\pi_1(X)$-covering $\tilde{M} \to M$ with map $\tilde{f}\colon \tilde{M} \to \tilde{X}$. Because f is a G-map, it is straightforward to pull back also the action of $\widetilde{G}$ to an action on $\tilde{M}$ which covers the action of G on M and such that $\tilde{f}$ is a $\widetilde{G}$-map (indeed, the $\widetilde{G}$-action on $\tilde{M}$ is defined mapping a point $(m, \tilde{x}) \in \tilde{M} \subset M \times \tilde{X}$ by $(\alpha, g) \in \widetilde{G}$ to $(gm, \alpha(\tilde{x}))$).

The same construction gives a bijection between G-invariant metrics on M and $\widetilde{G}$-invariant metrics on $\tilde{M}$, and also works for bordisms. These constructions are clearly inverse to each other and preserve all additional structure and define the required bijections. □